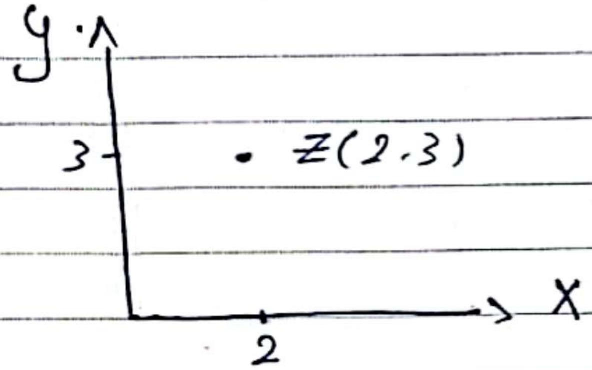
Comblex Aremental

$Z = x + iy$ $\quad i = \sqrt{-1}$ $\quad x, y \leftarrow R$

الأعداد الحقيقية

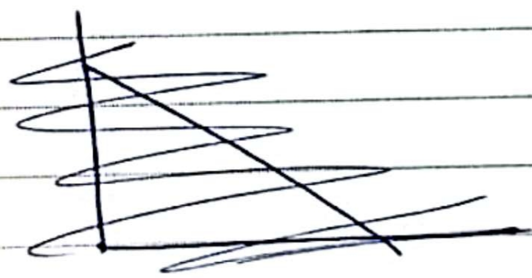


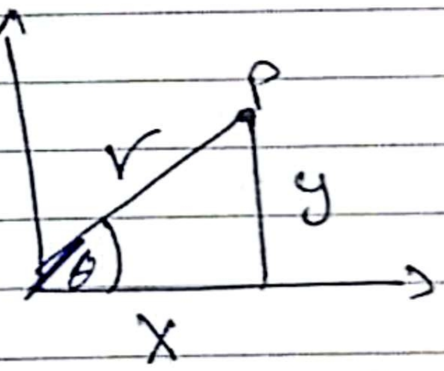
x = real Part
y = Injenary Part

$Z = 2 + 3i$

$z = 4 + 2i \qquad (\overset{x}{4}, \overset{y}{2})$

Polar form





$r = \sqrt{x^2 + y^2}$ $\quad \tan\theta = \frac{y}{x}$ $\quad \theta = \tan^{-1}$

$Z = r(\cos\theta + \sin\theta] = re^{i\theta}$

$y = r(co$
$X = rsi$

Note that $e^{i\theta} = \cos\theta + i\sin\theta$

find the value of

$$\left[\frac{2+3w}{2w^2+3}+\frac{5w^2-3}{5-3w}\right]^{30}$$

Solution

$$\frac{2+3w}{2w^2+3}=\frac{1}{w^2}\cdot\frac{2w^2+3w^3}{2w^2+3}$$

$$=\frac{1}{w^2}\cdot\frac{2w^2+3}{2w^2+3}=\frac{1}{w^2}=w$$

$$\frac{5w^2-3}{5-3w}=w^2\cdot\frac{5w^2-3}{5w^2-3w^3}=w^2$$

$$\cancel{\#}\quad=\left[w+w^2\right]^{30}\left[-1\right]^{30}=1$$

13/08/2025 21:13

if $z = 2[\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}]$

find z^3

$$z^3 = \cancel{z}\, 2^3[\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}]^3$$

$$= 2^3[\cos\pi + i\sin\pi]$$

$$2^3[-1] = -8$$

find the Cubic roots of 1

$z = 1 \begin{cases} x = 1 \\ y = 0 \end{cases}$ $\quad r = 1$

$\theta = \tan^{-1}\frac{y}{x} = 0.$

$$z = r[\cos\theta + \sin\theta] = [\cos 0 + i\sin 0]$$

$$= 1.$$

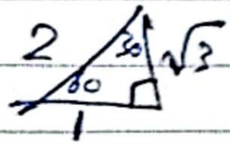
الأحد

if $Z_1 = r_1 [\cos\theta + i\sin\theta_2]$

$Z_2 = r_2 [\cos\theta + i\sin\theta_2]$

$Z_1 Z_2 = r_1 r_2 [\cos(\theta_1+\theta_2) + \sin(\theta_2+\theta_1)]$

$$\frac{Z_1}{Z_2} = \frac{r_1}{r_2} [\cos(\theta_1-\theta_2) + i\sin(\theta_1-\theta_2)]$$



$Z_1 = 2[\cos 60 + i\sin 60]$

$Z_2 = 4[\cos 30 + i\sin 30]$

find $Z_1 Z_2 = 8[\cos 90 + i\sin 90] = 8i$

$$\frac{Z_1}{Z_2} = 1/2 [\cos 30 + i\sin 30] = \frac{1}{2}\left(\frac{\sqrt{3}}{2} + i\right.$$

3 roots

$1, \ -\frac{1}{2}+\frac{\sqrt{3}}{2}, \ -\frac{1}{2}-\frac{\sqrt{3}}{2}$

$(1, \ w, \ w^2)$

Note that

$$1+w+w^2=0$$

$$w^3=1 \Rightarrow w^{3n}=1$$

$$w^4 = w \cdot w^3 = w$$

$$w^{52} = w^{1+17\cdot 3} = w \cdot w^{17\cdot 3} = w$$

If $\quad z_1 = 2+3i$

$z_2 = 2+5i$

find $z_1 + z_2$

$z_1 \cdot z_2$

$\frac{z_1}{z_2}$

Soulettion

$$z + z_2 = (2+3i) + (2-5i)$$

$$= 4 - 2i$$

$$z_1 z_2 = (2+3i)(2-5i)$$

$$= 19 - 4i$$

$$\frac{z_1}{z_2} = \frac{2+3i}{2-5i} \times \frac{2+5i}{2+5i} = \frac{(4-15)+i(16)}{29}$$

$$= \frac{1}{29}\left[-11 + 16i\right]$$

الأعتى

ouler $Z = r[\cos\theta + i\sin\theta]$

$$Z^n = r^n[\cos n\theta + i\sin n\theta]$$

$$Z^{1/n} = r^{1/n}[\cos\theta + \sin\theta]$$

$$Z^{1/n} = r^{1/n}\left[\cos\frac{\theta+2k\pi}{n} + i\sin\frac{\theta+2k\pi}{n}\right]$$

$$k = 0, 1, 2, 3, 4, 5, \ldots$$

$$Z = r(\cos\theta + i\sin\theta]$$

$$r = \sqrt{x^2+y^2} = |Z|$$

$$\tan\theta = \frac{y}{x} \qquad \pi < \theta < \pi$$

$Z = x + iy$ | $Z + Z' = 2x$

$Z = x - iy$ | $Z - Z' = Ziy$

Z' Consquit | $Z \cdot Z' = x^2 + y^2 = |Z|^2$

Ahmed Calil

الأحمد